PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Neuromuscular Electrical Stimulation for Children with Dysphagia: A Systematic Review
AUTHORS	Propp, Roni; Gill, Peter; Marcus, Sherna; Ren, Lily; Cohen, Eyal; Friedman, Jeremy; Mahant, Sanjay

VERSION 1 – REVIEW

REVIEWER	Hibberd , Judi Speech and Language Therapy Research Unit
REVIEW RETURNED	27-Sep-2021

GENERAL COMMENTS	Overall this is a very good piece with a few modifications which I can expand on. so the introduction contains inaccuracies that dont reflect on the outcomes but nevertheless need addressing so your content is accurate. Introduction: page 5 line 5- Occurs in about 1% of children is inaccurate and outdated. Look at the ASHA (American Speech Language Hearing Association) website. Put in Pedicatric Dysphagia and then refs and you will find its nearly 30 % and some say more depending on the disorder so cerebral palsy has something like 80%. page 5 line 5- ref 1 goes with above so 2008 is to old a ref to use in this context, its 13 years ago so more info available now.Use the ASHA site for an more up to date ref here. page 5 line 10 and 17- ref 3 is too old.Try this one instead. The Effect of Airway Aspiration on QOL of caregivers of children with cerebral palsy. S S Arslan Dysphagia Vol 33 June 2018 in the ESSD section page 5 line 21- ref 6 is too old, 1988 is 33 yrs ago. Try this one. Hewetson R, Singh S The lived in Experiences of mothers of children with chronic feeding and or swallowing difficulties. Dysphagia Vol 24 2009 322-332. This one is 12 yrs old but hasnt been superceded yet. or try Components of mealtime duration in a natural environment. Mishra et al but its an abstract from DRS in Dysphagia Vol 31 2016 page 5 line 39 and 40 reading compensatory strategies and oral motor exercises and ref 12.- This isnt entirely correct and should say, The current management of dysphagia consists of either compensations or therapeutic interventions. Compensations are equipment, position, pacing, taste, temperature, texture, environment and thickener. things that dont change the dysphagia just manage it allowing continuing feeding and drinking.

	Therapeutic interventions include oro motor work but also include chewing work, lip control and jaw control as well as a variety of tools including the IOPI and Iqoro and Abliex to name but a few.So NMES comes into a Therapeutic intervention. Your ref 12 only talks about oromotor. Try these instead or as well as 1. Caty Lazarus 2017 Dysphagia , History of the Use of and Impact of Compensatory Strategies in the Management of swallowing disorders 2. M Walshe , C Harding. Rehabilitation of Dysphagia in Children, Evidence and Directions for research 2015 Dysphagia Vol 30 page 6 line 3 - a new proposed approach. Is it though ?, the ref you are using is 2009 and is a good one, but 2009 isnt a new approach, its been around for 12 years and used extensively in the US and Europe though not the UK. call it something else, an approach that is gaining in acceptance perhaps. page 6 line 12- change oral motor exercises page 6 line 50- dysfunction in oral and or pharyngeal phases of swallowing. i would take issue with this , i use NMES all the time and it never impacts on the oral stage, it cant if its placed on submentals , its only going to impact on tongue base retraction, but this is a personal observation. page 16 line 17- change new proposed intervention page 20 references- change refs 1, 3 and 6 page 38/39/40/41- having just completed a systematic review myself this info isnt necessary. its to much for the reader and they dont care about this part. page 44- not this page either once the changes are made to the intorduction its good.

REVIEWER	Sandage , Mary J. Auburn University, Department of Communication Disorders
REVIEW RETURNED	29-Sep-2021

GENERAL COMMENTS	Review bmjopen-2021-055124 I have a PhD in applied Exercise Science, Kinesiology, I have practiced as a speech-language pathologist treating children with craniofacial disorders and upper airway disorders for almost 30 years, I teach anatomy and physiology for swallowing function through the lifespan, I have conducted several systematic reviews using the PRISMA guidelines, and I have authored one tutorial on use of NMES with infants and young children as well as other systematic reviews. I am qualified to review this manuscript. This is a well-written systematic review that adhered closely to the PRISMA guidelines and addressed a little studied topic of clinical importance. Abstract: 1) the baseline year that the search strategy employed for the search should be stated explicitly. Use of the “from inception” is too vague and lacks the specificity required for this systematic review to be replicated. Introduction:

	1) The prevalence of dysphagia in children is not well characterized by pooling all children under the age of 18 together into a single cohort (~ 1% of children) given that the prevalence in neonates is as high as 37% (Edwards, 2019). Suggest revising the initial paragraph to more clearly characterize the heterogeneity of this disorder by age and etiology. Including more detail will bolster the importance of this review. 2) P. 6, line 3, NMES is not a “new” approach. One of the studies referenced was from 2000, indicating that the data were collected over 20 years ago – not new. 3) P. 6, lines 37-44 – neuromuscular development in infants (which continues well past age 2 years) and the goal of habilitation in infants and young children are additional important differences from adult study of use of NMES – rehabilitation of lost skills– this should be acknowledged in this section because these are critical factors that should be considered for implementation of this modality. Methods: 1) P. 7, line 23-24 – the baseline year that the search strategy employed for the search should be stated explicitly. Use of the “from inception” is too vague and lacks the specificity required for this systematic review to be replicated. 2) In the Search Strategy (Appendix A); provide a rationale for inclusion of the following search terms in CINAHL as they do not seem to be directly related to the systematic review goals: (MH "Child Abuse, Sexual") OR (MH "Child Behavior Disorders+") OR (MH "Child, Medically Fragile") OR (MH "Child Day Care") OR (MH "Child Passenger Safety") OR (MH "Child Abuse+") OR (MH "Child Custody") OR (MH "Child Behavior Checklist") 3) P. 7, line 26, The search strategy was developed and implemented by a reference librarian? It is unclear if the clinical service providers or individuals who use NMES in clinic or research participated in the development of the search terms. Please clarify author involvement. If no authors were involved in determination of the search terms, this should be described as a limitation in the Discussion section. 4) The search strategy was enhanced by inclusion of papers other than those in only English. 5) The study selection procedures adhere to PRISMA guidelines. 6) Data extraction and management procedures adhered to PRISMA guidelines. 7) Risk of bias was addressed adhering to PRISMA guidelines. 8) The Gao, et al 2018 paper described use of acupuncture as the intervention of interest and evaluated the outcome measures for both study groups against use of acupuncture vs. no acupuncture following NMES + oral function training. It is difficulty to assert that the NMES contributed to the study findings given that it was not studied specifically. This claim/conclusion should be either removed or the verbiage qualified to frame improvements in function that were measured from this perspective. This limitation should be addressed for all references to this study in the Results section. For example, on p. 13, lines 8-10, large improvements in swallowing effectiveness are described without providing this essential context. It would be difficulty for any researcher to assert that the improvements observed were due to NMES, given that it was provided along with oral function training in both study groups, and the improvements described were in the NMES + oral function training + acupuncture group – indicating that the group differences quantified were more likely to be secondary to the

	addition of the acupuncture modality, not NMES. This is a mischaracterization of the findings. Subsequent claim made regarding improvements as described in the Gao, et al paper should also be clarified or removed from the Results and Discussion section. Given that the study aim was to quantify salivation improvements from the addition of acupuncture to NMES + oral function training, it is unclear how this study made the inclusion criteria for this systematic review. Suggest removing it from the papers included in the review. 9) The Marcus, et al 2019 paper – attributing swallowing improvements to use of NMES for the infants and toddlers consented for this study overlooked the probable maturational development during the course of the 2-4 month NMES protocol in these very young participants that may have accounted for some or all of the improvements documented. Attributing improvements to use of NMES, which is the stated goal of this systematic review is overstating the study findings. 10) Given the heterogeneity of the studies included and the influence of concurrent swallowing modalities in many of the studies, the standardized mean differences described in the Results section and the material in Figure 2 is hard to make sense of. Suggest removing this material from the Results section. Discussion: 1) P. 16, lines 14-17 – the authors' claim that "currently, there is no effective treatment that targets the underlying swallowing disorder in children with dysphagia" is not evidence supported. Suggest removing this sentence from the manuscript. 2) P. 16, lines17-19, remove the claim that NMES is a "new proposed intervention." With references dating back to 2000, this is clearly not a new modality. 3) The physiological rationale for use and implementation of NMES for children with acquired dysphagia, after having already developed effective and safe swallowing skills, is radically distinct from use and implementation for neonates and infants who need to habilitate swallowing function with a still-developing neuromuscular system. Given this, it is a bit hard to draw conclusions about the effectiveness for this modality with the heterogeneity of the study populations that comprises neonates through age 18, with both developmental and acquired dysphagia. The discussion section should include further exploration of the methodological implications of applying this modality to a developing neuromuscular system, as with the neonate or infant, versus to a disordered neuromuscular system (children with CP for example) versus a fully developed neuromuscular system (children after TBI or stroke). There are critical differences in the physiology of the infant group from the older child group that have implications for application of this modality, the most important of which is standardization of probe placement, frequency and duration (dose) of the modality, with and without accompanying exercises, positioning, or food/liquid texture changes. The neuromuscular implications of use of this modality with infants is described in: Epperson, H. E., & Sandage, M. J. (2019). Neuromuscular development in neonates and postnatal infants: implications for neuromuscular electrical stimulation therapy for dysphagia. Journal of Speech, Language, and Hearing Research, 62(8), 2575-2583. The recommendations section on p. 17, last paragraph, should include discussion of the importance of establishing standards for

	NMES implementation regarding probe placement, duration & frequency for future study of this modality. Basic research, perhaps with a juvenile animal model, should be conducted to establish NMES modality standards before further research with this modality is pursued. The physiological impacts of use of this modality on the developing neuromuscular system are unknown and it is of concern that is it being pursued in infants with methodologies determined largely from adult studies by clinicians with little to no training in electrophysiology– children are not little adults as the authors are aware, but the readers of this manuscript should it be published may not be aware. The authors of the systematic review should also devote more time in the Discussion qualifying the study outcomes from perspectives of concurrent modalities used when no control group was included and from an infant motor development perspective. Again, readers of this manuscript may not be as well versed on this modality with respect to use of concurrent swallowing exercise, positioning, or texture changes and they may be misled into thinking that NMES alone contributed to improvements documented in the studies reviewed. Without inclusion of these considerations, the study findings that NMES was responsible for swallowing improvement may be overstated. Given that the NMES modality is grossly understudied in children, particularly infants, the authors have a particularly important responsibility to accurately frame the study findings through careful analysis of each study's results. There should always be an acknowledgment in the Discussion of the limitation of the systematic review search terms – it is always likely that some papers were missed because of the study methods despite the authors' best effort. Additionally, the potential bias of inclusion of one of the author's (SM) publication in the review and data extraction conducted by the author (SM) needs to be acknowledged. I don't think that this is a flaw, but it would be best to acknowledge. Further investigations into the empirical benefits of the NMES modality for both children and adults are needed. It isn't surprising that so few papers made the final cut.

REVIEWER	Alamir, Abayneh Mekelle University, Department of Physiotherapy
REVIEW RETURNED	10-Oct-2021

GENERAL COMMENTS	I kindly appreciate your effort to conduct this manuscript. Result section - You were included articles done on other neurological impairment. Thus, how could you generalize this to the effect of NMES on dysphagia related with CP ? # discussion part - It lacks to meet your research question and too shallow . - please discuss it very well with the existing evidence

VERSION 1 – AUTHOR RESPONSE

Reviewer 1 Comments
Dr. Judi Hibberd, Speech and Language Therapy Research Unit

R1.1 Overall this is a very good piece with a few modifications which I can expand on so the introduction contains inaccuracies that don't reflect on the outcomes but nevertheless need addressing so your content is accurate.

Response: Thank you for providing these helpful comments to improve our manuscript.
Introduction

R1.2 page 5 line 5- Occurs in about 1% of children is inaccurate and outdated. Look at the ASHA (American Speech Language Hearing Association) website. Put in Pediatric Dysphagia and then refs and you will find its nearly 30 % and some say more depending on the disorder so cerebral palsy has something like 80%.

Response: Thank you for the comment. We have kept the reference to 1% of children, which was also referenced on the ASHA website. However, we have further qualified these findings and added estimates of prevalence for different subgroups. The Introduction now states,
"A United States national health survey conducted in 2012 found that amongst 61 million children about 1% suffered a swallowing problem lasting greater than 1 week in the previous year. Dysphagia is more prevalent in specific populations, including premature infants, and children with cerebral palsy, traumatic brain injury, cranio-facial abnormalities, and children with medical complexity. The prevalence of dysphagia ranges from 58%-99% in children with cerebral palsy and 10.5% in premature infants (<37 weeks)" (page 5, lines 5-17).

R1.3 page 5 line 5- ref 1 goes with above so 2008 is to old a ref to use in this context, its 13 years ago so more info available now. Use the ASHA site for an more up to date ref here.

Response: We replaced the reference with an updated review paper (Lawlor et al 2020) (page 5, line 5).

R1.4 page 5 line 10 and 17- ref 3 is too old. Try this one instead. The Effect of Airway Aspiration on QOL of caregivers of children with cerebral palsy. S S Arslan Dysphagia Vol 33 June 2018 in the ESSD section

Response: We replaced the reference with an updated review paper (Lawlor et al 2020) (page 5, line 14 and 23). We did not add the Arslan 2018 reference as it is a conference abstract.

R1.5 page 5 line 21- ref 6 is too old, 1988 is 33 yrs ago. Try this one. Hewetson R, Singh S The lived in Experiences of mothers of children with chronic feeding and or swallowing difficulties. Dysphagia Vol 24 2009 322-332. This one is 12 yrs old but hasn't been superceded Yet. or try Components of mealtime duration in a natural environment. Mishra et al but its an abstract from DRS in Dysphagia Vol 31 2016

Response: Thank you for the suggestion. The paper by Hewestson & Singh was very interesting, and we have cited the paper in replace of the previous reference (page 5, line 26).

R1.6 page 5 line 39 and 40 reading compensatory strategies and oral motor exercises and ref 12.- This isnt entirely correct and should say, The current management of dysphagia consists of either compensations or therapeutic interventions. Compensations are equipment, position, pacing, taste, temperature, texture, environment and thickener. things that dont change the dysphagia just manage it allowing continuing feeding and drinking. Therapeutic interventions include oro motor work but also include chewing work, lip control and jaw control as well as a variety of tools including the IOPI and Iqoro and Abliex to name but a few. So NMES comes into a Therapeutic intervention. Your ref 12 only talks about oromotor. Try these instead or as well as:
1. Caty Lazarus 2017 Dysphagia, History of the Use of and Impact of Compensatory Strategies in the Management of swallowing disorders
2. M Walshe, C Harding. Rehabilitation of Dysphagia in Children, Evidence and Directions for research 2015 Dysphagia Vol 30

Response: Thank you for the comment. We made changes to reflect the compensatory and therapeutic management of dysphagia and we added a more comprehensive list of compensatory strategies. The section now states: “The current management for dysphagia in children includes compensatory strategies and therapeutic interventions. Specifically, compensatory strategies include modifying diet (e.g., altering textures, thickening fluids, changing taste or temperature), postural changes, pacing, using different equipment (e.g., different spoon size), and/or changing environments” (page 5, lines 43-51). In addition, we have broadened the scope of therapeutic interventions included. The section now states: “Therapeutic interventions include oral motor exercises, which involve lip control and tongue control, and can be completed with the aid of a therapist and different equipment” (page 5, lines 50-55) We have not referred to the tools IOPI, Igoro, and Abilex as these are commercial products. We added the Lazarus et al 2017 reference (page 5, line 50). We did not add the Walshe & Harding 2015 reference as it is a conference abstract.

R1.7 page 6 line 3 - a new proposed approach. Is it though ?, the ref you are using is 2009 and is a good one, but 2009 isnt a new approach, its been around for 12 years and used extensively in the US and Europe though not the UK. call it something else, an approach that is gaining in acceptance perhaps.

Response: We agree – NMES is not a new approach, but a proposed approach that is starting to gain acceptance. However, it is not yet routine in clinical practice worldwide, as you have stated. For this reason, we have changed the phrasing as follows: “A proposed approach used to treat dysphagia is neuromuscular electrical stimulation (NMES)” (page 6, line 10).

R1.8 page 6 line 12- change oral motor exercises

Response: We have replaced “oral motor exercises” with “therapeutic interventions.” It now states, “Transcutaneous stimulation, administered by occupational therapists and/or speech language pathologists, is then combined with traditional swallowing therapy (e.g., compensatory strategies, therapeutic interventions, etc.)” (page 6, lines 12-20).

R1.9 page 6 line 50- dysfunction in oral and or pharyngeal phases of swallowing. I would take issue with this, I use NMES all the time and it never impacts on the oral stage, it can’t if its placed on submentals, its only going to impact on tongue base retraction, but this is a personal observation.

Response: We agree – NMES does not impact the oral phase but works primarily on the pharyngeal phase. The phrase “oral and/or pharyngeal phases of swallowing” was intended only as a definition of oropharyngeal dysphagia. However, to avoid any misinterpretation, we have removed the brackets. It now states, “Therefore, the objective of this systematic review is to assess the effectiveness of NMES for treatment of oropharyngeal dysphagia in children” (page 7, lines 3-6).

R1.10 page 16 line 17- change new proposed intervention

Response: We have changed the phrasing. It now states “NMES is a proposed intervention for the treatment of dysphagia” (page 16, line 16). Please see response to comment R1.7 for a more detailed explanation.

R1.11 page 20 references- change refs 1, 3 and 6

Response: We have changed these references where requested. Please see responses to comment R1.3, R1.4, and R1.5 for a more detailed explanation.

R1.12 page 38/39/40/41- having just completed a systematic review myself, this info isn’t necessary. Its to much for the reader and they don’t care about this part.

Response: Thank you for your comment. To comply with the PRISMA guidelines, we are including the search strategy. However, we have included the search strategies in an appendix, and therefore, this does not have to be reviewed by readers who are not interested in this section.

R1.13 page 44- not this page either

Response: Thank you for your comment. To comply with the PRISMA guidelines, we are including the search strategy. However, we have included the search strategies in an appendix, and therefore, this does not have to be reviewed by readers who are not interested in this section.

R1.14 once the changes are made to the introduction its good.

Response: Thank you.

Reviewer 2 Comments
Dr. Mary J. Sandage, Auburn University

R2.1 This is a well-written systematic review that adhered closely to the PRISMA guidelines and addressed a little studied topic of clinical importance.

Response: Thank you for the detailed and helpful comments on improving our manuscript.

Abstract
R2.2 The baseline year that the search strategy employed for the search should be stated explicitly. Use of the “from inception” is too vague and lacks the specificity required for this systematic review to be replicated.

Response: Thank you for the comment. We agree that it increases specificity allowing for the systematic review to be replicated. We have explicitly added the baseline year for each of the databases. The abstract now states, “A search was performed on November 2020 in MEDLINE (from 1946), EMBASE (from 1947), PsycINFO (from 1806), CINAHL (from 1937), CENTRAL (from 1996) and Scopus (from 1970) databases were searched from inception to November 2020” (page 3, lines 13-15).

R2.3 The prevalence of dysphagia in children is not well characterized by pooling all children under the age of 18 together into a single cohort (~ 1% of children) given that the prevalence in neonates is

as high as 37% (Edwards, 2019). Suggest revising the initial paragraph to more clearly characterize the heterogeneity of this disorder by age and etiology. Including more detail will bolster the importance of this review.

Response: Thank you for the comment. We agree – we have revised the introduction to include more detail on the prevalence of dysphagia by etiology and age. Specifically, we added the following sentence: “The prevalence of dysphagia ranges from 58%-99% in children with cerebral palsy and 10.5% in premature infants (<37 weeks)” (page 5, lines 14-18).

R2.4 P. 6, line 3, NMES is not a “new” approach. One of the studies referenced was from 2000, indicating that the data were collected over 20 years ago – not new.

Response: We agree – NMES is not a new approach, but it is also not yet routine in clinical practice. For this reason, we have changed the phrasing as follows: “A proposed approach used to treat dysphagia is neuromuscular electrical stimulation (NMES)” (page 6, line 10).

R2.5 P. 6, lines 37-44 – neuromuscular development in infants (which continues well past age 2 years) and the goal of habilitation in infants and young children are additional important differences from adult study of use of NMES – rehabilitation of lost skills– this should be acknowledged in this section because these are critical factors that should be considered for implementation of this modality.

Response: We agree – the goal of rehabilitation is an important difference to highlight children and adults. It now states, “that the etiology of dysphagia in adults is very different (i.e. oropharyngeal malignancies, stroke, dementia) and that the goal of rehabilitation is different in adults (i.e., adults are undergoing rehabilitation for lost skills and children are undergoing rehabilitation to develop skills). In addition, neuroplasticity changes with age” (page 6, lines 43-51).

Methods
R2.6 P. 7, line 23-24 – the baseline year that the search strategy employed for the search should be stated explicitly. Use of the “from inception” is too vague and lacks the specificity required for this systematic review to be replicated.

Response: Thank you for your comment. The methods section now states “MEDLINE (OVID from 1946), EMBASE (OVID from 1947), PsycINFO (OVID from 1806), CINAHL (EBCSO from 1937), Cochrane Central Register of Controlled Trials (OVID from 1996) and Scopus (from 1970) databases were searched from inception to July 19, 2019 and updated on November 26, 2020” (page 6, lines 32-39). Please see response to comment R2.2 for a more detailed explanation.

R2.7 In the Search Strategy (Appendix A); provide a rationale for inclusion of the following search terms in CINAHL as they do not seem to be directly related to the systematic review goals:
(MH "Child Abuse, Sexual") OR (MH "Child Behavior Disorders+") OR (MH "Child, Medically Fragile") OR (MH "Child Day Care") OR (MH "Child Passenger Safety") OR (MH "Child Abuse+") OR (MH "Child Custody") OR (MH "Child Behavior Checklist")

Response: Though these terms do not seem to be directly related to the systematic review goals, they are part of the Pediatric search string. The pediatric search hedges were developed by the University of Alberta, which we have cited in our paper. This search string was then adapted for each of the databases. Specifically, the pediatric search string for CINAHL was developed by Sandy Campbell. The following is the citation:
Campbell, Sandy. Filter to Retrieve Pediatric Articles in the EBSCO CINAHL Database. John W. Scott Health Sciences Library, University of Alberta. Rev. April

18, 2020 https://docs.google.com/document/d/1HBmwNb0NXPVBudA_w8UsjfkJdF73p9fWROuX_OX8-8s/edit

R2.8 P. 7, line 26, The search strategy was developed and implemented by a reference librarian? It is unclear if the clinical service providers or individuals who use NMES in clinic or research participated in the development of the search terms. Please clarify author involvement. If no authors were involved in determination of the search terms, this should be described as a limitation in the Discussion section.

Response: The search strategy was developed and implemented by a reference librarian. This is noted in the Methods section “The search strategy was developed and carried out by LR, a reference librarian with expertise in conducting systematic reviews” (page 7, lines 39-42).

In addition, authors were involved in the development of the search terms (specifically, RP, SM, SM, and PG). Of these, SM, SM, and PG are clinicians involved in the care of children with dysphagia. We have revised the Methods section to include these details. It now states that “RP, SM, PG, SM were also involved in the development of the search terms. Three of the five authors are clinicians involved in the care of children with dysphagia (SM, SM, PG)” (page 7, lines 41-46).

R2.9 The search strategy was enhanced by inclusion of papers other than those in only English.

Response: Thank you.

R2.10 The study selection procedures adhere to PRISMA guidelines.

Response: Thank you.

R2.11 Data extraction and management procedures adhered to PRISMA guidelines.

Response: Thank you.

R2.12 Risk of bias was addressed adhering to PRISMA guidelines.

Response: Thank you.

R2.13 The Gao, et al 2018 paper described use of acupuncture as the intervention of interest and evaluated the outcome measures for both study groups against use of acupuncture vs. no acupuncture following NMES + oral function training. It is difficulty to assert that the NMES contributed to the study findings given that it was not studied specifically. This claim/conclusion should be either removed or the verbiage qualified to frame improvements in function that were measured from this perspective. This limitation should be addressed for all references to this study in the Results section. For example, on p. 13, lines 8-10, large improvements in swallowing effectiveness are described without providing this essential context. It would be difficulty for any researcher to assert that the improvements observed were due to NMES, given that it was provided along with oral function training in both study groups, and the improvements described were in the NMES + oral function training + acupuncture group – indicating that the group differences quantified were more likely to be secondary to the addition of the acupuncture modality, not NMES. This is a mischaracterization of the findings. Subsequent claim made regarding improvements as described in the Gao, et al paper should also be clarified or removed from the Results and Discussion section.
Given that the study aim was to quantify salivation improvements from the addition of acupuncture to NMES + oral function training, it is unclear how this study made the inclusion criteria for this systematic review. Suggest removing it from the papers included in the review.

Response: Thank you for your comment. As you have suggested, given the use of co-interventions, we agree that it is difficult to conclude that the observed effect was due to NMES in the Goa et al 2018 paper. Therefore, we have now excluded the treatment group (NMES + oral motor training + acupuncture) from our analysis. Instead, we are only including the control group in our analysis (NMES + oral motor training). As you have suggested, this poses several limitations. First, there is no control group, and therefore, we are unable to conclude the effect of NMES separately from the effect of oral motor training. In addition, there is the effect of maturational development with time. These limitations are true of all included studies without a control group, and therefore, we have further elaborated on these limitations in the discussion. The discussion now states,

"Second, we included a wide range of studies, several of which did not have a control group. In these cases, it cannot be determined that the observed effect was due to NMES. Some of these studies had additional intervention components, such as oral motor exercises, and this may have contributed to the observed effect. In addition, there is a natural maturational development that occurs with time, which may have also contributed to the observed effect in other studies. However, when there is limited evidence, it is important to survey the literature more broadly, and therefore, we included these studies in our review" (page 17, lines 37-51).

Finally, though there are analytical difficulties in the Goa et al 2018 paper (as discussed above), we have included the paper in our systematic review as it met our inclusion/exclusion criteria – it is a study of children diagnosed with oropharyngeal dysphagia by clinical assessment.

R2.14 The Marcus, et al 2019 paper – attributing swallowing improvements to use of NMES for the infants and toddlers consented for this study overlooked the probable maturational development during the course of the 2-4 month NMES protocol in these very young participants that may have accounted for some or all of the improvements documented. Attributing improvements to use of NMES, which is the stated goal of this systematic review is overstating the study findings.
Response: We agree – there is a maturational development that may have contributed to the observed in the Marcus et al 2019 study. We have further elaborated on this limitation in the discussion. It now states, "Second, we included a wide range of studies, several of which did not have a control group. In these cases, it cannot be determined that the observed effect was due to NMES… In addition, there is a natural maturational development that occurs with time, which may have also contributed to the observed effect" (page 17, lines 37-48). The lack of control group is also stated as a limitation of the study in the Marcus et al 2019 paper. Studies with control groups are necessary to definitively draw conclusions of the effectiveness of NMES, which supports our conclusion that high-quality trials are needed.

R2.15 Given the heterogeneity of the studies included and the influence of concurrent swallowing modalities in many of the studies, the standardized mean differences described in the Results section and the material in Figure 2 is hard to make sense of. Suggest removing this material from the Results section.

Response: Thank you for the comment. We agree – due to similar concerns, we did not calculate a pooled intervention effect or conduct a meta-analysis. However, we do think it is important to have an overall summary of the studies, which we have conducted using standardized mean differences (described in the results section and displayed graphically in Figure 2). Standardized mean differences were only calculated for the RCTs and the control groups of each study was used as the reference group. Of the four studies analyzed as RCTs, two studies used the same oral motor exercises in both groups and sham/placebo NMES. One study only used NMES in the treatment group (i.e., there were no additional intervention component). In the last study, there was

a difference in the exercises using in the treatment and control groups, which poses a limitation. We have outlined the limitations of these studies in the manuscript (page 17, line 34 - page 19, line 7).

Discussion
R2.16 P. 16, lines 14-17 – the authors' claim that "currently, there is no effective treatment that targets the underlying swallowing disorder in children with dysphagia" is not evidence supported. Suggest removing this sentence from the manuscript.

Response: Thank you for the comment. We have removed this sentence from the manuscript.

R2.17 P. 16, lines17-19, remove the claim that NMES is a "new proposed intervention." With references dating back to 2000, this is clearly not a new modality.

Response: We have changed the phrasing. It now states "NMES is a proposed intervention for the treatment of dysphagia" (page 16, line 16). Please see response to comment R2.4 for a more detailed explanation.

R2.18 The physiological rationale for use and implementation of NMES for children with acquired dysphagia, after having already developed effective and safe swallowing skills, is radically distinct from use and implementation for neonates and infants who need to habilitate swallowing function with a still-developing neuromuscular system. Given this, it is a bit hard to draw conclusions about the effectiveness for this modality with the heterogeneity of the study populations that comprises neonates through age 18, with both developmental and acquired dysphagia. The discussion section should include further exploration of the methodological implications of applying this modality to a developing neuromuscular system, as with the neonate or infant, versus to a disordered neuromuscular system (children with CP for example) versus a fully developed neuromuscular system (children after TBI or stroke). There are critical differences in the physiology of the infant group from the older child group that have implications for application of this modality, the most important of which is standardization of probe placement, frequency and duration (dose) of the modality, with and without accompanying exercises, positioning, or food/liquid texture changes. The neuromuscular implications of use of this modality with infants is described in:
Epperson, H. E., & Sandage, M. J. (2019). Neuromuscular development in neonates and postnatal infants: implications for neuromuscular electrical stimulation therapy for dysphagia. Journal of Speech, Language, and Hearing Research, 62(8), 2575-2583.

Response: Thank you for your comment. We agree – there is heterogeneity of the study populations, which has implications for the use of NMES. As you have suggested, there is large differences between neonates and infants with a developing muscular system, children with a disordered neuromuscular system (such as children with cerebral palsy), and children with a developed neuromuscular system (such as acquired dysphagia due to TBI, for example). We have addressed this issue from several perspectives in our systematic review.

First, we have explained the heterogenous population of children with dysphagia in the "Findings in the Context of Previous Research" section of the discussion. It now states, "In addition, the population of children with dysphagia is heterogenous, including neonates and infants with a developing muscular system, children with a disordered neuromuscular system (such as children with cerebral palsy), and children with a developed neuromuscular system (such as children with acquired dysphagia)" (page 16, lines 48-55).

Second, we have specified the limitations in interpreting our systematic review in the "Limitations" section of the discussion. It now states, "Third, the populations of children with dysphagia in the included studies are clinically heterogenous – neonates and infants, children with neuromuscular

disorders, and children with acquired dysphagia. The populations different physiologically, which has implications for use of NMES" (page 17, line 55 – page 29, line 8).

Third, we have highlighted recommendations for future studies in the "Implications for Future Research" section of the discussion. It states, "In addition, the effects of the intervention should be assessed in different patient populations (e.g., young infants and older children) and aetiologies (e.g., premature infants, neurologic impairment)" (page 18, lines 43-48).

Finally, we have not calculated a pooled intervention effect or conducted a meta-analysis due to the clinical heterogeneity of the populations (page 17, lines 34-37).

Thank you for including the reference to Epperson & Sandage et al 2019. It was an interesting paper and we have cited it in our paper (page 16, line 54).

R2.19 The recommendations section on p. 17, last paragraph, should include discussion of the importance of establishing standards for NMES implementation regarding probe placement, duration & frequency for future study of this modality. Basic research, perhaps with a juvenile animal model, should be conducted to establish NMES modality standards before further research with this modality is pursued. The physiological impacts of use of this modality on the developing neuromuscular system are unknown and it is of concern that is it being pursued in infants with methodologies determined largely from adult studies by clinicians with little to no training in electrophysiology– children are not little adults as the authors are aware, but the readers of this manuscript should it be published may not be aware.

Response: Thank you for the comment. We added the importance of establishing standards as a recommendation for future research in the "Implications for Future Research" section of the discussion. It now states, "It is also important to establish standards for the implementation of NMES regarding probe placement, duration, frequency, voltage, and current intensity" (page 18, line 52 - page 19, line 3). In addition, it is important to highlight the difference between adults and children regarding the use of NMES. This has been elaborated on in the Introduction and Discussion of the paper (page 6, lines 41-53; page 16, line 43).

R2.20 The authors of the systematic review should also devote more time in the Discussion qualifying the study outcomes from perspectives of concurrent modalities used when no control group was included and from an infant motor development perspective. Again, readers of this manuscript may not be as well versed on this modality with respect to use of concurrent swallowing exercise, positioning, or texture changes and they may be misled into thinking that NMES alone contributed to improvements documented in the studies reviewed. Without inclusion of these considerations, the study findings that NMES was responsible for swallowing improvement may be overstated. Given that the NMES modality is grossly understudied in children, particularly infants, the authors have a particularly important responsibility to accurately frame the study findings through careful analysis of each study's results.

Response: We agree – we previously commented on these issues in the Discussion, but we have now further elaborated on the limitations. The Limitations section now states, "Second, we included a wide range of studies, several of which did not have a control group. In these cases, it cannot be determined that the observed effect was due to NMES. Some of these studies had additional intervention components, such as oral motor exercises, and this may have contributed to the observed effect. In addition, there is a natural maturational development that occurs with time, which may have also contributed to the observed effect in other studies. However, when there is limited evidence, it is important to survey the literature more broadly, and therefore, we included these studies in our review" (page 17, lines 36-50).

R2.21 There should always be an acknowledgment in the Discussion of the limitation of the systematic review search terms – it is always likely that some papers were missed because of the study methods despite the authors' best effort.

Response: It is certainly possible that some studies may have been missed despite a comprehensive systematic search of the literature. We have added this as a limitation. It now states, "Finally, some studies may have been missed despite a comprehensive systematic search of the literature" (page 18, lines 7-10).

R2.22 Additionally, the potential bias of inclusion of one of the author's (SM) publication in the review and data extraction conducted by the author (SM) needs to be acknowledged. I don't think that this is a flaw, but it would be best to acknowledge.

Response: We have acknowledged that several of the study authors were involved in previous research on NMES, including one of the included studies in the review. We revised the Methods section, which now states, "Of note, several study authors who were involved in the systematic review (SM, JF, SM), have conducted research in this area and previously conducted a study on NMES" (page 7, lines 14-20). In addition, data was primarily extracted by RP and SM verified the extracted data. This is stated in the Data Extraction and Management section of the Methods (page 9, line 52).

R2.23 Further investigations into the empirical benefits of the NMES modality for both children and adults are needed. It isn't surprising that so few papers made the final cut.

Response: We agree – further research is needed to establish the evidence of NMES as a treatment for dysphagia. In the paper, we suggested that further studies are needed and outlined recommendations for future research (page 18, line 16 - page 19, line 8). We have not made any changes to the paper based on this comment.

Reviewer 3 Comments
Mr. Abayneh Alamir, Mekelle University

Results
R3.1 You were included articles done on other neurological impairment. Thus, how could you generalize this to the effect of NMES on dysphagia related with CP?

Response: Thank you for your comment. Our systematic review did not focus on children with cerebral palsy, rather it focused on children with dysphagia, including children with neurological impairments (e.g., children with cerebral palsy) and children without neurological impairments. The effect of NMES was not generalized to children with CP, rather we have elaborated on how the effect of NMES may vary in different populations. Please see response to comment R2.18 for further elaboration. We have not made any changes to the paper based on this comment.

Discussion
R3.2 It lacks to meet your research question and too shallow. Please discuss it very well with the existing evidence

Response: Thank you for your comment. Based on the editor and reviewers' comments, we have worked to improve the discussion. We have elaborated on several areas of the discussion section, including the context of previous research, recommendations for future studies, and limitations,

as well as including additional citations. Please see responses to comments R2.18, R2.19, R2.20, and R2.21 where we further elaborate on specific points of the Discussion.
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VERSION 2 – REVIEW

REVIEWER	Hibberd , Judi Speech and Language Therapy Research Unit
REVIEW RETURNED	14-Dec-2021

GENERAL COMMENTS	The authors have taken account of the all of the comments made and I thank them for that. It reads well and will be of interest to the wider professional audience. well done.

REVIEWER	Sandage , Mary J. Auburn University, Department of Communication Disorders
REVIEW RETURNED	16-Dec-2021

GENERAL COMMENTS	The authors carefully considered all reviewer concerns and suggestions with a much clarified manuscript as a result. All of this reviewer's concerns have been carefully addressed. There is only one word change that should be made prior to publication: On page 7, first paragraph, parenthetic example verbiage should indicate "children are undergoing habilitation to develop skills" NOT "undergoing rehabilitation to develop skills." Acquisition of a motor skill not yet learned is typically referred to as "habilitation" not "rehabilitation."

REVIEWER	Alamir, Abayneh Mekelle University, Department of Physiotherapy
REVIEW RETURNED	18-Dec-2021

GENERAL COMMENTS	The revised manuscript had good improvement.